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USN: 23BTRCT029

DATE: 10/10/2026

BATCH: 3

COLLEGE: JAIN UNIVERSITY

TITLE: AWS Cloud & Data Engineering

Mysql

1. Write a SQL query to display each department's ID, total number of employees, and average salary.

```
153
154 • SELECT department_id,
155        COUNT(*) AS employee_count,
156        AVG(salary) AS average_salary
157    FROM employees
158    GROUP BY department_id;
159
```

Output | Result #19

department_id	employee_count	average_salary
D001	3	74000.0000
D002	4	57250.0000
D003	4	78250.0000
D004	4	59500.0000

2. Department average salary above ₹70,000

```
159
160 • SELECT department_id,
161         AVG(salary) AS average_salary
162   FROM employees
163   GROUP BY department_id
164   HAVING AVG(salary) > 70000;
165
```

Output | Result #20

department_id	average_salary
D001	74000.0000
D003	78250.0000

3. Write a SQL query to calculate the total project cost assigned to each employee, displaying the employee ID and total cost.

```
181
182 • SELECT employee_id,
183         SUM(project_cost) AS total_cost
184   FROM projects
185   GROUP BY employee_id;
186
```

Output | Result #24

employee_id	total_cost
E001	850000
E002	420000
E003	675000
E004	380000
E005	510000
E006	720000
E007	460000
E008	325000
E009	590000
E010	575000

4. Write a SQL query to display the details of the five employees with the highest salaries, ordered from highest to lowest.

```sql
SELECT *
FROM employees
ORDER BY salary DESC
LIMIT 5;
```

Output — Result #25

employee_id	employee_name	department_id	salary	joining_date	city
E011	Karthik Raman	D003	88000	07/16/2018	Chennai
E005	Suresh Iyer	D001	82000	04/17/2019	Mumbai
E006	Kavya Reddy	D003	79000	07/25/2022	Hyderabad
E001	Arun Kumar	D003	75000	06/15/2021	Chennai
E010	Sneha Patel	D001	72000	03/29/2021	Ahmedabad

5. Write a SQL query to find the second-highest distinct salary among employees.

```sql
SELECT MAX(salary)
FROM employees
WHERE salary < (SELECT MAX(salary) FROM employees);
```

Output — Result #26

MAX(salary)
82000

6. Write a SQL query to display all employee details for employees whose salary is greater than the average salary of all employees

```sql
SELECT *
FROM employees
WHERE salary > (SELECT AVG(salary) FROM employees);
```

Output — Result #27

employee_id	employee_name	department_id	salary	joining_date	city
E001	Arun Kumar	D003	75000	06/15/2021	Chennai
E004	Divya Menon	D001	68000	11/08/2021	Kochi
E005	Suresh Iyer	D001	82000	04/17/2019	Mumbai
E006	Kavya Reddy	D003	79000	07/25/2022	Hyderabad
E010	Sneha Patel	D001	72000	03/29/2021	Ahmedabad
E011	Karthik Raman	D003	88000	07/16/2018	Chennai
E013	Vivek Nair	D003	71000	05/11/2020	Bengaluru

7. Write a SQL query to display each employee's ID, employee name, and total cost of their assigned projects by joining the `employees` and `projects` tables.

```sql
SELECT e.employee_id,
       e.employee_name,
       SUM(p.project_cost) AS total_cost
FROM employees e
JOIN projects p
ON e.employee_id = p.employee_id
GROUP BY e.employee_id, e.employee_name;
```

Output | Result #32

employee_id	employee_name	total_cost
E001	Arun Kumar	850000
E002	Priya Sharma	420000
E003	Rahul Verma	675000
E004	Divya Menon	380000
E005	Suresh Iyer	510000
E006	Kavya Reddy	720000
E007	Vignesh Raj	460000
E008	Ananya Singh	325000

8. Write a SQL query to display each customer's name along with the amount of their transaction by joining the `customers` and `transactions` tables.

```sql
SELECT c.customer_name,
       t.amount
FROM customers c
JOIN transactions t
ON c.customer_id = t.customer_id;
```

Output | Result #37

customer_name	amount
Ramesh Traders	45000
Lakshmi Stores	28000
Bharat Foods	67500
Meera Enterprises	52000
Krishna Textiles	38500
Asha Technologies	18000
Rajesh Motors	92000
Nandini Exports	41500
Vijay Agencies	73500
Shree Medicals	12500

9. Write a SQL query to find the department with the highest average employee salary. Display the department ID and average salary.

```sql
SELECT department_id,
       AVG(salary) AS avg_salary
FROM employees
GROUP BY department_id
ORDER BY avg_salary DESC
LIMIT 1;
```

Output | Result #39

department_id	avg_salary
D003	78250.0000

10. Write a SQL query to display each department's ID and minimum salary, considering only departments whose minimum salary is greater than ₹40,000.

```sql
SELECT department_id,
       MIN(salary)
FROM employees
GROUP BY department_id
HAVING MIN(salary) > 40000;
```

Output | Result #40

department_id	MIN(salary)
D001	68000
D002	51000
D003	71000
D004	55000

11. Write a SQL query to find the IDs of employees who are not assigned to any project.

```sql
SELECT e.employee_id
FROM employees e
LEFT JOIN projects p
ON e.employee_id = p.employee_id
WHERE p.project_id IS NULL;
```

Output | Result #44

employee_id
E014
E011
E013
E012
E015

12. Write a SQL query to display all details of the project with the highest project cost.

```sql
SELECT *
FROM projects
ORDER BY project_cost DESC
LIMIT 1;
```

Output | Result #45

project_id	project_name	employee_id	project_cost	start_date
P001	ERP Upgrade	E001	850000	01/15/2024

13. Write a SQL query to display each department's ID, department name, and employee count, including departments that have no employees.

```sql
SELECT d.department_id,
       d.department_name,
       COUNT(e.employee_id) AS employee_count
FROM departments d
LEFT JOIN employees e
ON d.department_id = e.department_id
GROUP BY d.department_id, d.department_name;
```

Output | Result #46

department_id	department_name	employee_count
D001	Finance	3
D002	HR	4
D003	IT	4
D004	Sales	4

14. Write a SQL query to display each employee's name, department name, and salary by joining the `employees` and `departments` tables

```sql
SELECT e.employee_name, d.department_name, e.salary
FROM employees e
INNER JOIN departments d
ON e.department_id = d.department_id;
```

Output | Result #21

employee_name	department_name	salary
Divya Menon	Finance	68000
Suresh Iyer	Finance	82000
Sneha Patel	Finance	72000
Priya Sharma	HR	62000
Ananya Singh	HR	51000
Mohammed Faiz	HR	57000
Pooja Deshmukh	HR	59000
Arun Kumar	IT	75000
Kavya Reddy	IT	79000
Karthik Raman	IT	88000
Vivek Nair	IT	71000
Rahul Verma	Sales	58000
Vignesh Rai	Sales	55000

15. Write a SQL query to calculate the salary range for each department. Display the department ID and salary range, where the salary range is the difference between the maximum and minimum salaries.

```sql
SELECT department_id,
       MAX(salary) - MIN(salary) AS salary_range
FROM employees
GROUP BY department_id;
```

Output | Result #47

department_id	salary_range
D001	14000
D002	11000
D003	17000
D004	9000

Linux commands

```
[ec2-user@ip-172-31-18-218 ~]$ mkdir aws
[ec2-user@ip-172-31-18-218 ~]$ cd aws
[ec2-user@ip-172-31-18-218 aws]$ touch ec2.txt
[ec2-user@ip-172-31-18-218 aws]$ echo "ec2 is a virtual remote server inside the aws" >ec2.txt
[ec2-user@ip-172-31-18-218 aws]$ nano ec2.txt
[ec2-user@ip-172-31-18-218 aws]$ ls
ec2.txt
[ec2-user@ip-172-31-18-218 aws]$ touch vpc.txt
[ec2-user@ip-172-31-18-218 aws]$ mv vpc.txt vpc_notes.txt
[ec2-user@ip-172-31-18-218 aws]$ rm vpc_notes.txt
[ec2-user@ip-172-31-18-218 aws]$ head ec2.txt
ec2 is a virtual remote server inside the aws
It is a service provided by aws and a resource of aws
ec2 instance given by aws is auto scalable
[ec2-user@ip-172-31-18-218 aws]$ pwd
/home/ec2-user/aws
[ec2-user@ip-172-31-18-218 aws]$ chown user aws.txt
chown: invalid user: ‘user’
[ec2-user@ip-172-31-18-218 aws]$ df -h
Filesystem        Size  Used Avail Use% Mounted on
devtmpfs          4.0M     0  4.0M   0% /dev
tmpfs             457M     0  457M   0% /dev/shm
tmpfs             183M  444K  183M   1% /run
efivarfs          128K  2.7K  121K   3% /sys/firmware/efi/efivars
/dev/nvme0n1p1    8.0G  1.6G  6.4G  20% /
tmpfs             457M     0  457M   0% /tmp
/dev/nvme0n1p128   10M  1.3M  8.7M  13% /boot/efi
tmpfs              92M     0   92M   0% /run/user/1000
```

```
[ec2-user@ip-172-31-18-218 aws]$ grep "aws" ec2.txt
ec2 is a virtual remote server inside the aws
It is a service provided by aws and a resource of aws
ec2 instance given by aws is auto scalable
[ec2-user@ip-172-31-18-218 aws]$ find ec2
find: ‘ec2’: No such file or directory
[ec2-user@ip-172-31-18-218 aws]$ find . -type f -name "*.txt"
./ec2.txt
[ec2-user@ip-172-31-18-218 aws]$ whoami
ec2-user
[ec2-user@ip-172-31-18-218 aws]$ cat ec2.txt
ec2 is a virtual remote server inside the aws
It is a service provided by aws and a resource of aws
ec2 instance given by aws is auto scalable
[ec2-user@ip-172-31-18-218 aws]$ free -h
               total        used        free      shared  buff/cache   available
Mem:           913Mi       155Mi       511Mi       0.0Ki       246Mi       626Mi
Swap:             0B          0B          0B
[ec2-user@ip-172-31-18-218 aws]$ cd ..
```

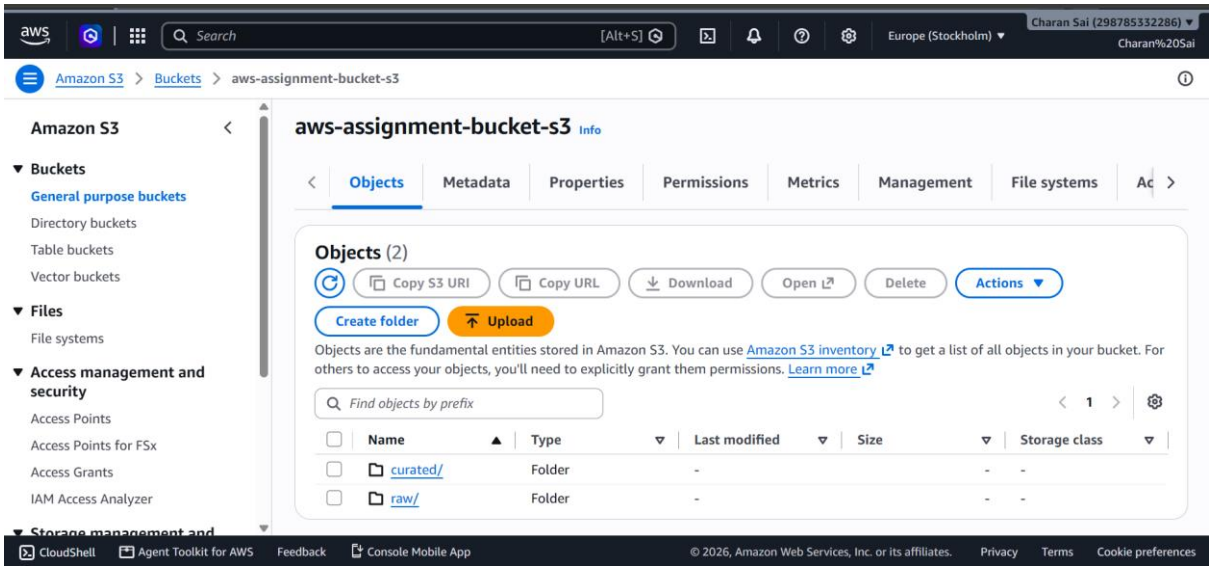
```
[ec2-user@ip-172-31-18-218 ~]$ ps aux
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.0  1.9 173928 17836 ?        Ss   06:09   0:00 /usr/lib/systemd/systemd --switched-root --system --deserialize=32
root           2  0.0  0.0      0     0 ?        S    06:09   0:00 [kthreadd]
root           3  0.0  0.0      0     0 ?        S    06:09   0:00 [pool_workqueue_release]
root           4  0.0  0.0      0     0 ?        I<   06:09   0:00 [kworker/R-rcu_gp]
root           5  0.0  0.0      0     0 ?        I<   06:09   0:00 [kworker/R-sync_wq]
root           6  0.0  0.0      0     0 ?        I<   06:09   0:00 [kworker/R-kvfree_rcu_reclaim]
root           7  0.0  0.0      0     0 ?        I<   06:09   0:00 [kworker/R-slub_flushwq]
root           8  0.0  0.0      0     0 ?        I<   06:09   0:00 [kworker/R-netns]
root          10  0.0  0.0      0     0 ?        I<   06:09   0:00 [kworker/0:0H-events_highpri]
root          11  0.0  0.0      0     0 ?        I    06:09   0:00 [kworker/0:1-events_power_efficient]
root          13  0.0  0.0      0     0 ?        I<   06:09   0:00 [kworker/R-mm_percpu_wq]
root          14  0.0  0.0      0     0 ?        S    06:09   0:00 [ksoftirqd/0]
root          15  0.0  0.0      0     0 ?        I    06:09   0:00 [rcu_preempt]
root          16  0.0  0.0      0     0 ?        S    06:09   0:00 [rcu_exp_par_gp_kthread_worker/0]
root          17  0.0  0.0      0     0 ?        S    06:09   0:00 [rcu_exp_gp_kthread_worker]
root          18  0.0  0.0      0     0 ?        S    06:09   0:00 [migration/0]
root          19  0.0  0.0      0     0 ?        S    06:09   0:00 [cpuhp/0]
root          20  0.0  0.0      0     0 ?        S    06:09   0:00 [cpuhp/1]
root          21  0.0  0.0      0     0 ?        S    06:09   0:00 [migration/1]
root          22  0.0  0.0      0     0 ?        S    06:09   0:00 [ksoftirqd/1]
root          24  0.0  0.0      0     0 ?        I<   06:09   0:00 [kworker/1:0H-events_highpri]
root          27  0.0  0.0      0     0 ?        S    06:09   0:00 [kdevtmpfs]
root          28  0.0  0.0      0     0 ?        I<   06:09   0:00 [kworker/R-inet_frag_wq]
root          29  0.0  0.0      0     0 ?        I    06:09   0:00 [rcu_tasks_kthread]
root          30  0.0  0.0      0     0 ?        I    06:09   0:00 [rcu_tasks_rude_kthread]
root          31  0.0  0.0      0     0 ?        I    06:09   0:00 [rcu_tasks_trace_kthread]
```

```
[ec2-user@ip-172-31-18-218 ~]$ ls
aws
[ec2-user@ip-172-31-18-218 ~]$ cd aws
[ec2-user@ip-172-31-18-218 aws]$ chmod 755 ec2.txt
```

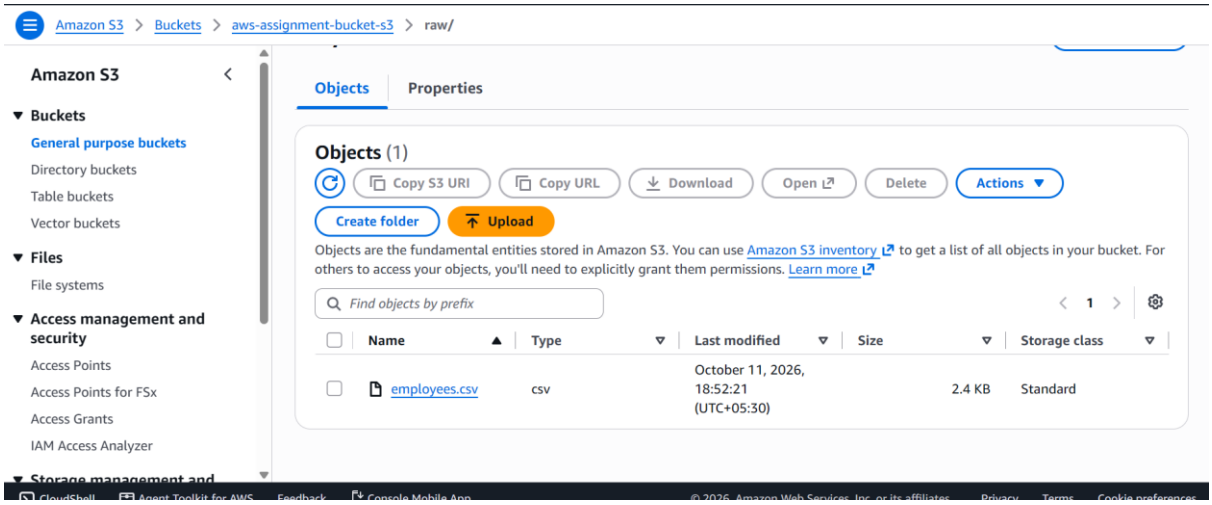
Amazon S3

- S3 bucket created

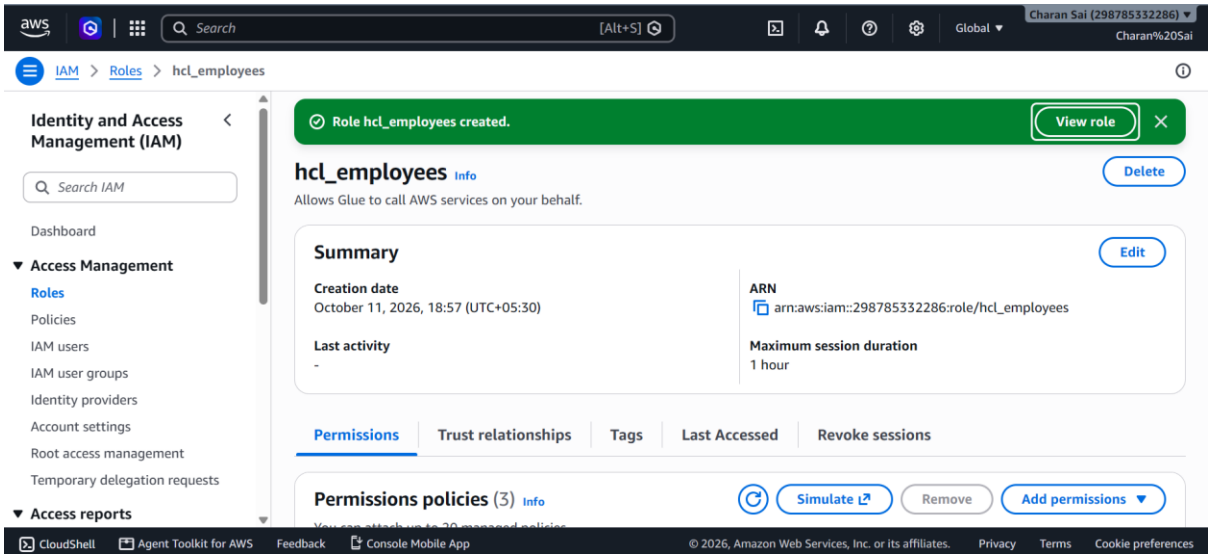
- raw/ folder
- curated/ folder



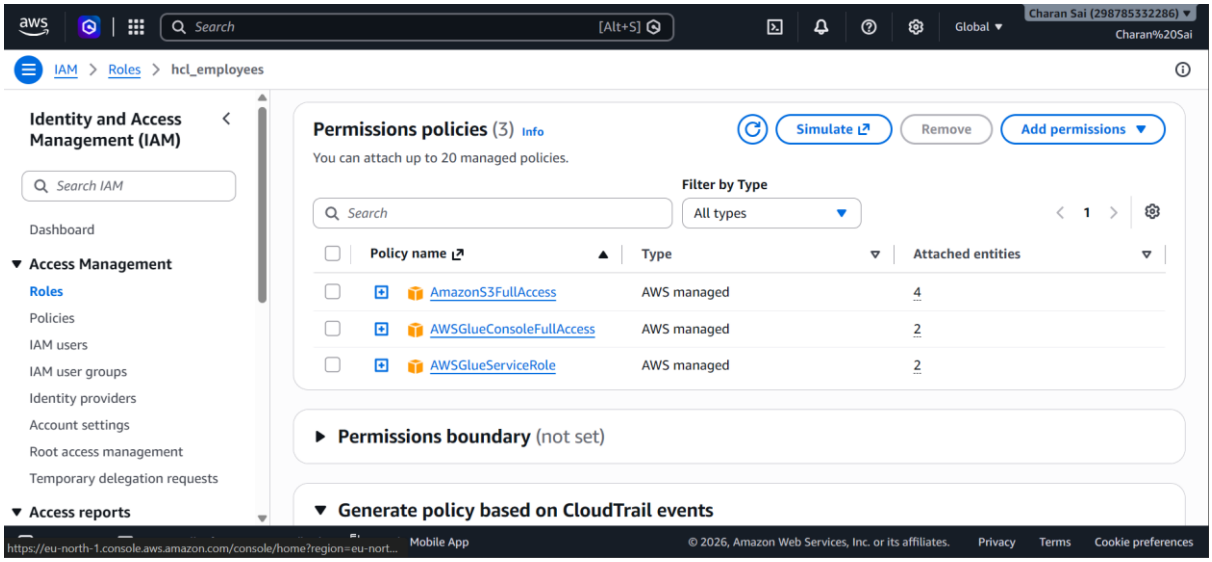
employees.csv uploaded inside raw/

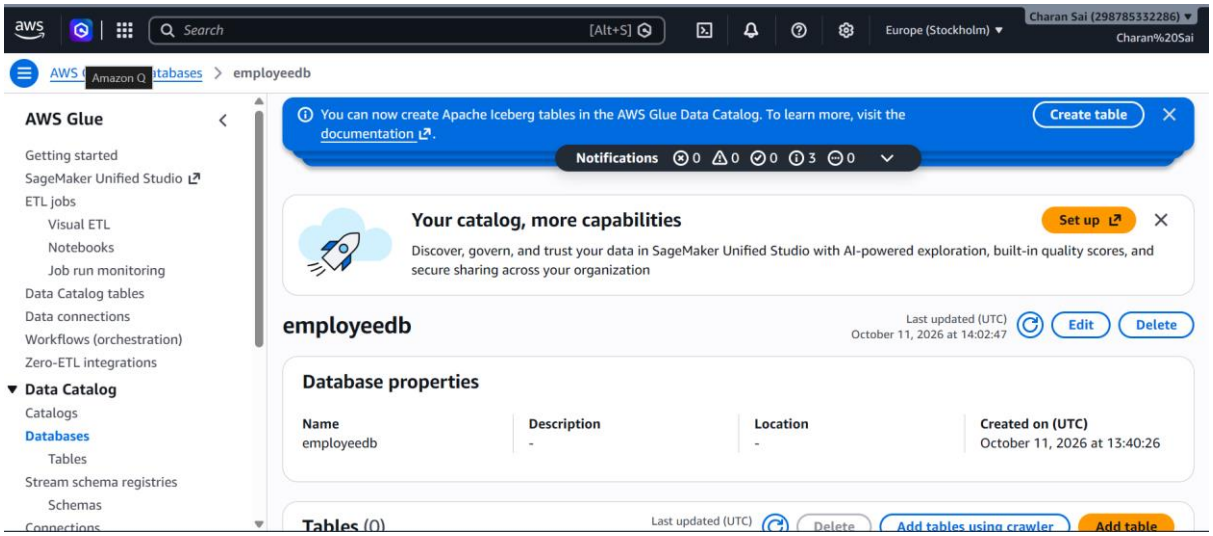


IAM Role used for AWS Glue

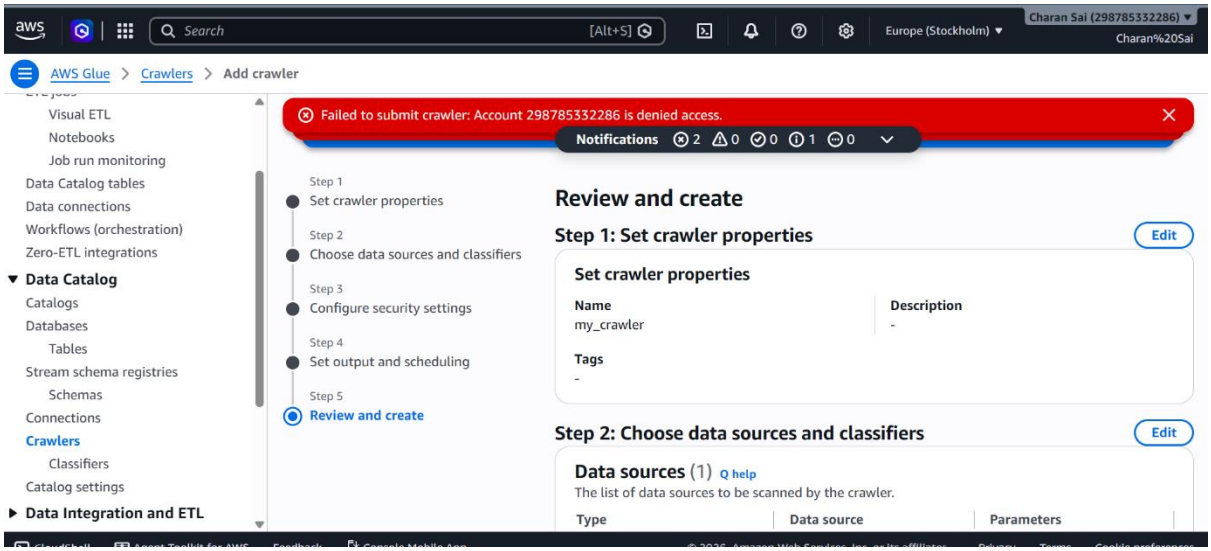


Permissions policies attached to the created role



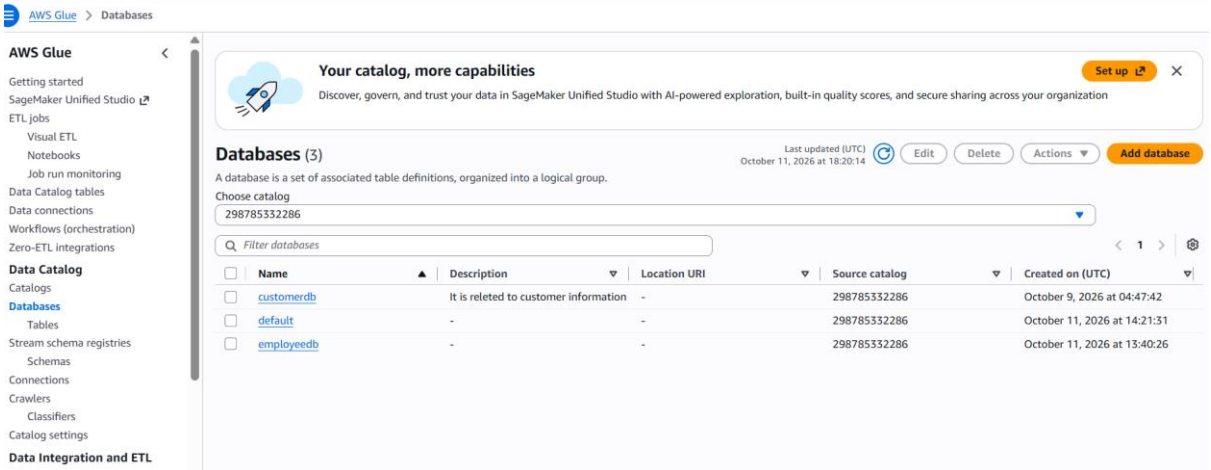


Glue Crawler

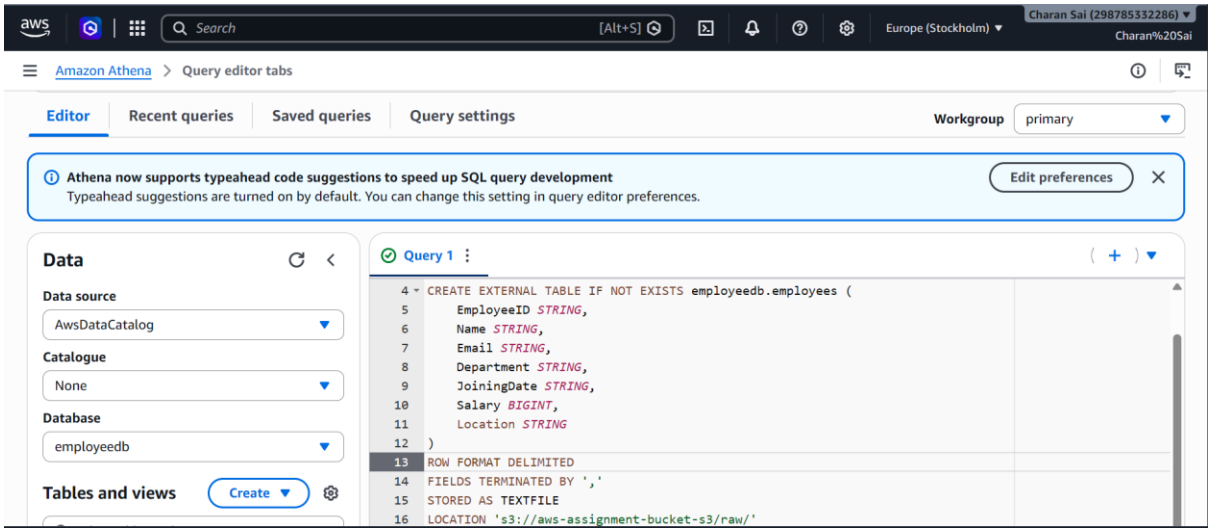


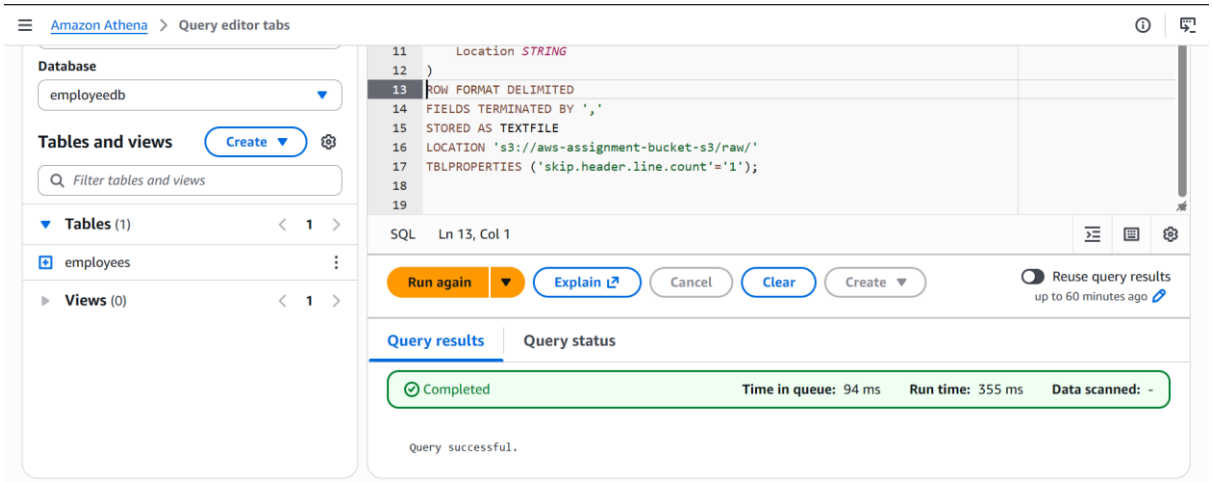
Glue Data Catalog

1.Employeedb has been created successfully

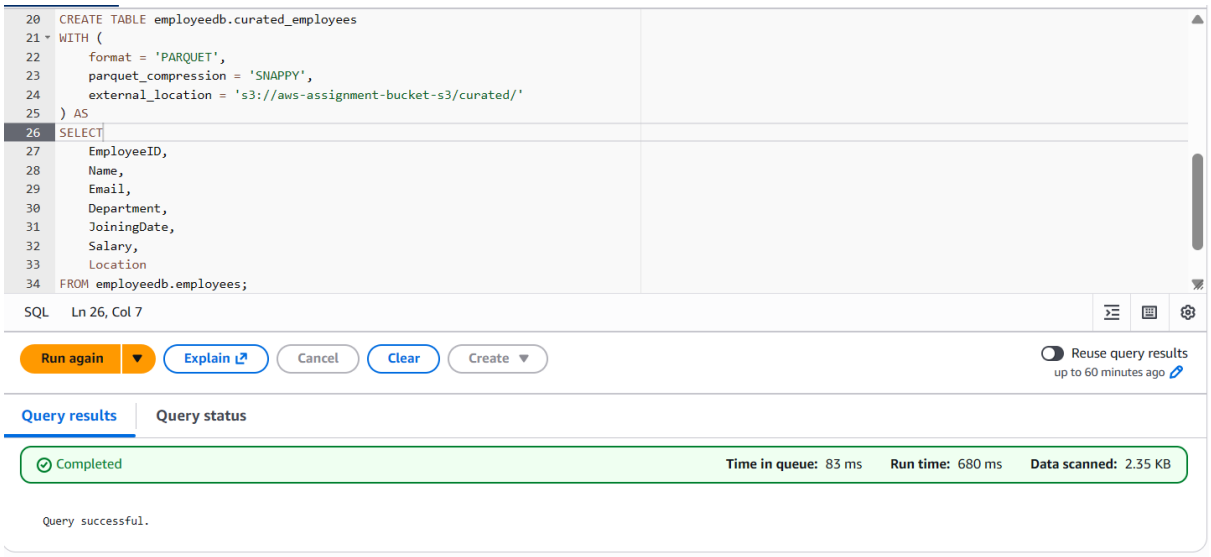


Employees Table is successfully created





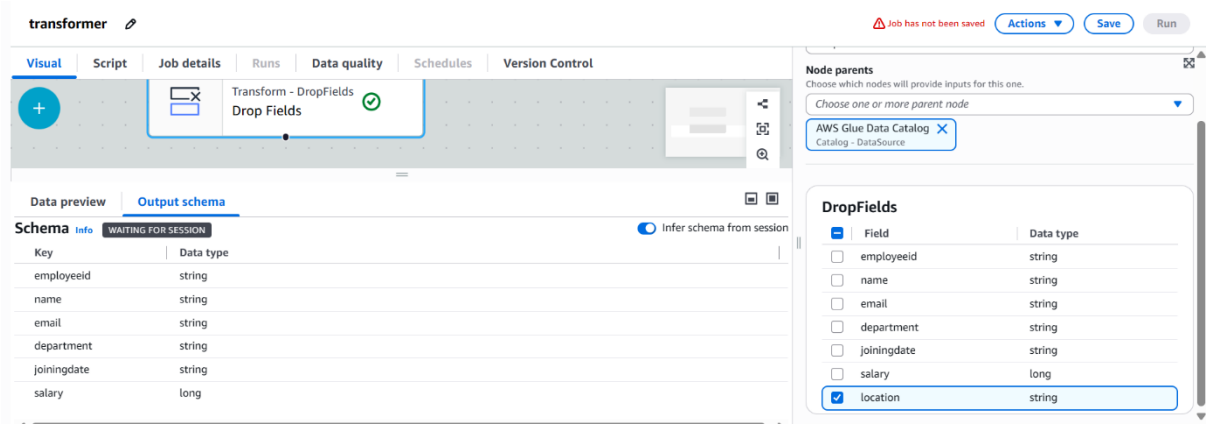
Curated employees table is created successfully



Glue Visual ETL

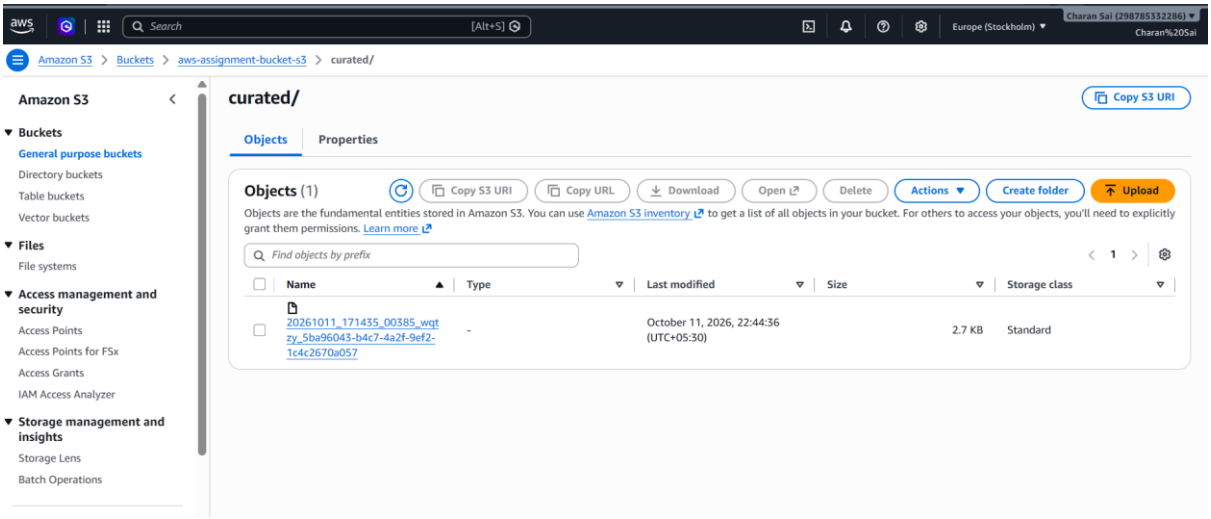
Glue Transformation

• Data fields are transformed



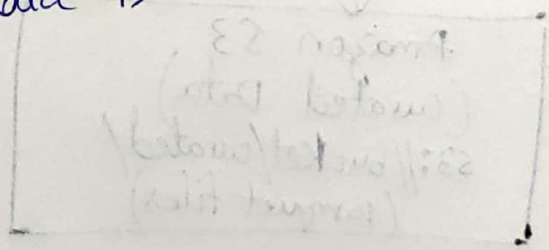
Final S3 Output

• curated/ folder • Generated output files • Parquet output



Execution workflow

1. First, I uploaded the employee CSV file to an S3 bucket where the raw employee data is stored.
2. I used an AWS Glue Crawler to scan the CSV file and identify its columns and datatypes.
3. I also created "IAM" role to manage the permissions to a specific identify for performing Glue visual ETL, Glue Crawler and Glue Data Catalog operations.
4. In the ETL workflow, I selected the wanted columns and dropped field(column such as location) and prepared the data for further use.
5. I converted the processed data from CSV format to Parquet format with Snappy compression.
6. In the end, The transformed data is stored in a separate curated/ folder which was created along with raw folder in S3 bucket, Both raw data and transformed data are stored separately in their respective folders.
7. Raw Data in the CSV file format is extracted, transformed and loaded into S3 bucket curated folder through ETL process.
8. Finally, curated data is saved into curated folder successfully.



Scanned with OKEN Scanner

AWS Data transformation & Conversion Process flow

